

Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding

Hundman et al. (2018) - KDD '18, NASA Jet Propulsion Laboratory

1. Overview

This paper, published at KDD 2018, was authored by researchers at NASA's Jet Propulsion Laboratory. It addresses the challenge of automatically detecting anomalies in spacecraft telemetry data — the streams of sensor readings transmitted back to Earth from satellites and rovers.

The two spacecraft datasets used are:

- SMAP (Soil Moisture Active Passive) — an Earth-orbiting satellite
- MSL (Mars Science Laboratory) — the Curiosity rover on Mars

The paper proposes two main technical contributions: (1) using LSTM neural networks to predict expected telemetry values, and (2) a novel nonparametric dynamic thresholding approach to decide whether prediction errors indicate a real anomaly.

2. The Problem Being Solved

Spacecraft have thousands of telemetry channels measuring temperature, radiation, power, and computational activity. Monitoring all of these manually is impractical. Existing systems had serious limitations:

- **Simple fixed thresholds. Miss anomalies that occur within normal value ranges or that have a temporal/contextual component.** Out-of-Limits (OOL) alarms:
- **High recall but low precision — too many false positives that waste engineer time.** Expert systems & nearest-neighbor methods:
- **Only use four statistical features; cannot capture temporal patterns.** Clustering methods (e.g., IMS):
- **Only applied to a handful of channels due to on-board memory limits.** Neural network approaches (e.g., ELMER):

The authors argue that all existing approaches fail to scale and fail to detect contextual anomalies — readings that look normal in isolation but are wrong given the surrounding context. This is where LSTMs provide a major advantage.

3. Types of Anomalies

Anomaly Type	Definition	Detectable by OOL?
Point	A single value falls far outside the normal distribution of values.	Yes, if threshold is well-set
Contextual	A value looks normal globally, but is anomalous given surrounding time-series context (e.g., temperature spike during a specific command sequence).	No — requires temporal modeling
Collective	A sequence of values is anomalous as a group, even if each individual value appears normal.	No — requires sequence modeling

In the paper, contextual and collective anomalies are merged into a single “contextual” category because both require temporal context to detect. The dataset contains 59% point anomalies and 41% contextual anomalies.

4. The SMAP/MSL Dataset

The labeled anomaly data was derived from Incident Surprise, Anomaly (ISA) reports — official NASA documents filed when a spacecraft experiences an unexpected event.

Metric	SMAP	MSL	Total
Total anomaly sequences	69	36	105
Point anomalies	43 (62%)	19 (53%)	62 (59%)
Contextual anomalies	26 (38%)	17 (47%)	43 (41%)
Unique telemetry channels	55	27	82
Unique ISAs	28	19	47
Telemetry values evaluated	429,735	66,709	496,444

Telemetry channels include not just raw sensor values but also one-hot encoded command data — information about which module received a command and whether it was sent or received. This command context is critical for detecting contextual anomalies.

5. LSTM-Based Telemetry Prediction

5.1 Why LSTMs?

LSTMs (Long Short-Term Memory networks) are a type of Recurrent Neural Network (RNN) specifically designed to learn long-range temporal dependencies. They solve the vanishing gradient problem that plagued earlier RNNs through a gating mechanism:

- **Decides which past information to discard**Forget gate:
- **Decides which new information to store**Input gate:
- **Decides what to pass to the next time step**Output gate:

For anomaly detection, the LSTM is trained on normal (non-anomalous) data to learn what “normal” behavior looks like. When it encounters an anomaly, its predictions will be significantly wrong, producing large prediction errors.

5.2 Single-Channel Architecture

A key design decision: one separate LSTM model is trained per telemetry channel. This provides:

- Channel-level traceability — you can identify which exact sensor is anomalous
- Independent control — early stopping and tuning per channel
- Scalability — channels with simpler behavior need less capacity

5.3 Model Architecture & Parameters

Parameter	Value	Rationale
Hidden layers	2	Sufficient capacity without overfitting
Units per layer	80	Balances model size and prediction accuracy
Sequence length (ls)	250	Captures long-term temporal patterns
Training iterations	35	With early stopping to prevent overfitting
Dropout	0.3	Regularization to reduce overfitting
Batch size	64	Standard for time-series training
Optimizer	Adam	Adaptive learning rate
Input dimensions	25 (SMAP), 55 (MSL)	Different command module sets

The models achieved an average normalized absolute prediction error of 5.9% (5.5% for SMAP, 6.8% for MSL). The higher error for MSL reflects that Curiosity performs a much more diverse range of activities compared to SMAP’s routine orbits.

6. Nonparametric Dynamic Thresholding

6.1 The Problem with Gaussian Thresholds

The most common approach is to model prediction errors as a Gaussian (normal) distribution and flag values beyond a certain number of standard deviations. This fails because spacecraft telemetry errors are empirically non-Gaussian — the paper verifies this using the D’Agostino-Pearson normality test and rejects the Gaussian assumption for all channels at $\alpha = 0.005$.

6.2 The Proposed Method

The method works in three steps:

- **Step 1 — Compute & Smooth Errors:** Prediction error $e(t) = |\text{actual} - \text{predicted}|$. Errors are smoothed using Exponentially Weighted Moving Average (EWMA) to dampen noise spikes.
- **Step 2 — Find the Best Threshold:** Test thresholds $\epsilon = \mu(es) + z\sigma(es)$ for z in $\{2.5, 3.0, \dots, 10.0\}$. Choose the ϵ that maximizes the percent decrease in mean and standard deviation of errors when anomalous points are removed — without generating too many anomalous sequences.
- **Step 3 — Score Anomalies:** Each anomalous sequence gets a severity score based on how far its peak error exceeds the chosen threshold, normalized by the mean and standard deviation of all smoothed errors.

The key insight is that this threshold selection is fully unsupervised — it requires no labeled data and makes no parametric distributional assumptions.

7. Mitigating False Positives

7.1 Anomaly Pruning

After thresholding, a pruning step eliminates borderline anomalous sequences. The procedure:

- Sort anomalous sequences by their peak smoothed error in descending order
- Step through consecutive pairs and compute the percent decrease
- If the decrease between two consecutive anomalies falls below a minimum threshold p (set to 0.13), all lower-scoring sequences are reclassified as nominal

Effect of pruning: overall recall decreases by only 4.8 percentage points (84.8% $\rightarrow$ 80.0%), while precision increases by 38.6 percentage points (48.9% $\rightarrow$ 87.5%). This is a massive win for deployment.

7.2 Learning from History

Once anomaly history accumulates, a minimum score $s_{\min}$ can be set per channel. Future anomalies below $s_{\min}$ are reclassified as nominal. This handles recurring normal spacecraft behaviors (e.g., periodic maintenance commands) that would otherwise keep triggering false alarms. User-provided labels can also set $s_{\min}$ directly.

8. Experimental Results

All methods use the same LSTM predictions. The only difference is in the thresholding approach used to classify prediction errors.

Thresholding Approach	Precision	Recall	F0.5 Score
Non-Parametric + Pruning ($p=0.13$) — BEST	87.5%	80.0%	0.71
Non-Parametric without Pruning ($p=0$)	48.9%	84.8%	0.47
Gaussian Tail ($\epsilon_{\text{norm}}=0.0001$)	87.5%	66.7%	0.66
Gaussian Tail ($\epsilon_{\text{norm}}=0.01$)	75.8%	71.4%	0.62
Gaussian Tail + Pruning ($\epsilon_{\text{norm}}=0.01$, $p=0.13$)	91.7%	62.9%	0.66

F0.5 weights precision twice as heavily as recall, reflecting the operational reality that false positives burn engineer time and erode trust in the system.

Recall by anomaly type (best method):

Spacecraft	Recall — Point	Recall — Contextual
MSL (Curiosity rover)	78.9%	58.8%
SMAP (satellite)	95.3%	76.0%
Overall	90.3%	69.0%

Contextual anomaly recall (69.0%) is lower than point anomaly recall (90.3%), which is expected — these are inherently harder to detect. The fact that the LSTM-based approach catches 69% of contextual anomalies is a significant achievement compared to any OOL method, which would catch close to 0%.

9. Relevance to the CERT-SAT Project

This paper is the primary reference for the LSTM-based anomaly detection component in CERT-SAT. Key takeaways for implementation:

- **2-layer LSTM with 80 units per layer, dropout 0.3, Adam optimizer, sequence length 250, batch size 64.**Architecture:
- **Telemetry values + one-hot encoded command information. For CERT-SAT, include any available command/mode metadata.**Input features:
- **Use the nonparametric method (z from 2.5 to 10.0, $p = 0.13$) rather than a fixed Gaussian threshold. This directly addresses non-stationarity in the SMAP/MSL data.**Thresholding:
- **F0.5 score (precision-weighted) is preferred for operational settings. Also track PR-AUC and Point-Adjusted F1.**Evaluation metric:
- **Implement anomaly pruning (p parameter). Later, use s_{min} thresholds once anomaly history builds up.**False positive strategy:

- **Train one model per channel for traceability. This also allows early stopping to be applied independently.**Single-channel models:
- **Isolation Forest is used as the baseline in CERT-SAT. Expect LSTM to outperform significantly on contextual anomalies.**Baseline comparison:

The open-source Telemanom implementation (github.com/khundman/telemanom) provides a reference codebase that can be studied and adapted for CERT-SAT.

10. Key Terms Glossary

Term	Meaning
SMAP	Soil Moisture Active Passive — NASA Earth satellite, one of the two datasets
MSL	Mars Science Laboratory — the Curiosity rover, second dataset
ISA Report	Incident Surprise, Anomaly — official NASA report of unexpected spacecraft events
OOL	Out-of-Limits — simple fixed threshold alarm, the baseline industry standard
EWMA	Exponentially Weighted Moving Average — used to smooth prediction errors
Telemanom	Open-source implementation of this paper (GitHub: khundman/telemanom)
F0.5 score	Harmonic mean weighting precision twice as much as recall
Nonparametric	Does not assume a specific probability distribution (e.g., Gaussian) for the data
Pruning (anomalies)	Post-hoc step to remove borderline anomalous sequences with similar scores to nominal data
smin	Minimum anomaly score threshold set per channel to suppress recurring false positives